

ules like opencv2, espeak, PiCamera, gpiozero, pytesseract, tesseract-ocr, and numpy are used. As all of these are open source software, they can be installed easily through the following commands on any Raspberry Pi computer using Linux.

```
# sudo apt-get update
# sudo apt-get upgrade
# sudo pip3 install pillow
# sudo pip3 install pytesseract
# sudo apt-get install tesseract-ocr
# sudo pip3 install opencv-python
# sudo pip3 install numpy
# sudo apt-get install espeak
# sudo apt install python3-gpiozero
# sudo pip3 install gpiozero
# sudo apt-get install python3-picamera
```

With the Python code created, open the camera and check the vehicle's number plate. First, the HCSR04 checks whether the vehicle is within four to six metres of the camera or not. If not, the system will ask

the driver to bring the vehicle closer, within the required distance.

Once the vehicle is in place, the camera takes a picture of the front of the vehicle and checks whether it is indeed a vehicle. If it finds it indeed to be a vehicle, it zooms on its number plate and analyses the number plate. It removes characters such as '-' and ' ' from the number and displays it on the system. The database of the known number plates is kept ready in a string array [carpool] separated by commas. Each number element of the database is compared against this number to find out the match.

Our database being a small one, is managed in the following one-string array, called carpool, which can hold up to 200 numbers comfortably:

```
carpool = ['OR06F4630','UP16BY0538','MP10CA4768','MH03AW2684','MHO3AW2684']
```

If a match is found, it will trigger the GPIO21 pin to open the gate or close it. There are many more GPIO pins available, which can be used for multiple triggering of the boom barrier or the gate mechanism—like green light, red light, gate open, gate close, etc.

During actual deployment, the author found that for a cheap quality camera, the lighting creates an issue. What is clear during day hours becomes unclear during nighttime.

However, with good-quality camera the problem is overcome. In his prototype, the author used a camera that had IR light source attached, which gave an extra edge during nighttime detection.

A 5-metre camera extension cord was used for the prototype, so that the camera could be placed at the centre of the boom barrier for proper detection.



Fig. 7: Car and number plate being detected

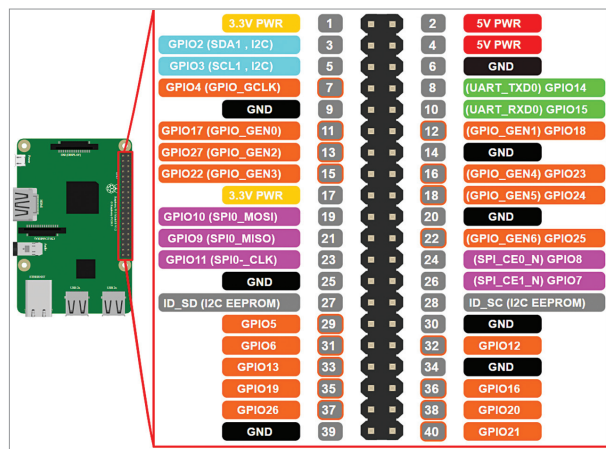


Fig. 8: Pins of Raspberry pi

## EFY Note

The source code of this project is available for free download at [www.electronicsforu.com](http://www.electronicsforu.com)

The TFT touchscreen, the keyboard, mouse, etc are all optional. Once the project is deployed for an actual operation, all these can be stripped off the computer.

Raspberry Pi 4 or Pi 3 can be used for this project. The speaker is to be inserted in its earphone jack. An amplifier or Bluetooth Amplibox can be used for louder and clear voice reproduction. The espeak synthesizer has many options, such as "espeak", "-ven + f3", "abcde fgh 1234567890"

"- ven + f3" is for female voice reproduction. You can search the internet for more options. The Python source code's snippet is shown in Fig. 6.

## Testing

To test the system, run the code after connecting power to the Raspberry Pi. The car with the number plate shown in Fig. 7 is one of the several cars with which the author tested his prototype. Each car was asked to slow down between four and metres from the camera.

The prototype was able to aptly identify the car and its number plate and compared it with carpool array to trigger the GPIO-21 of the Raspberry Pi computer. But sometimes it mistook 0 (zero) for O (O for onion). This can be tackled in logical level, such that the letters at 3,4,7,8,9,10 positions are considered to be only and the rest as alphabets. The main Python script automatic-numberplate identification.py runs non-stop.

Fig. 8 shows pins' description of the Raspberry Pi used during testing. **EFY**



Somnath Bera, General Manager at NTPC, is an enthusiast of IoT and electronics